

The stationary solution: the actual algorithm

Guess prices, solve households, aggregate, update prices, repeat

What “stationary” means here. Prices and aggregates are constant; households still age, save and have children. Hold economic parameters fixed. The current model has one market, four-year periods, 17 age cells and 120 wealth nodes. States also record inherited tenure/product, income, lifetime births and children at home.

1. **Guess the house price p . Compute rent and household resources.** With a constant price, no-arbitrage gives $r = (R - 1 + \delta + \tau_H)p$, where R is the gross financial return, δ depreciation and τ_H the property-tax rate, all in four-year units. Given p and r , construct each state’s income, sale proceeds, housing costs, borrowing/down-payment limits and feasible choices. The retained historical baseline fixes transfers at zero.
2. **Start at the oldest age; solve conditional household problems.** The final age uses the terminal bequest payoff. Then move one age backward at a time. At each state, consider no birth and an additional birth, and each feasible rental/owner product. For each branch and product, choose saving and current consumption/housing using the already computed next-age values. Denote this optimized deterministic value by $Q_{d,k}$, where $d \in \{0, 1\}$ is the birth outcome and k the housing product.
For a renter at given saving, split resources net of family consumption/housing requirements using Cobb–Douglas shares; cap rental housing and put the remainder into consumption. For an owner product, housing is fixed and the budget gives consumption. Only a one-dimensional saving search remains. For saving, search every feasible piecewise-linear continuation segment: compare endpoints, rental-cap kinks and interior first-order-condition solutions. This is exhaustive for the numerical approximation. Income risk, survival, child departures and bequests enter continuation values. Store policies and values; **repeat for every state at that age.**
3. **Combine these values in the sequential choice operator.** First integrate the *subsequent* housing shocks conditional on the birth outcome. The housing inclusive value is:

$$H_d = \kappa \log \sum_k \exp(Q_{d,k}/\kappa).$$

Here κ is the housing-choice taste scale. H_d is the expected optimized value of housing and saving after outcome d . Now construct the values of waiting, W , and attempting, A , before biological success is known:

$$W = H_0, \quad A = (1 - \pi)H_0 + \pi(H_1 - C_1), \quad P_{\text{attempt}} = \frac{e^{A/\sigma}}{e^{W/\sigma} + e^{A/\sigma}}.$$

The biological success probability is π . The utility cost C_1 applies only to a first birth; it is zero for later births. The fertility taste scale σ can differ between first and later attempts. Birth probability is πP_{attempt} . The pre-choice value passed to the preceding age is:

$$V = \sigma \log\{e^{W/\sigma} + e^{A/\sigma}\}.$$

Economic timing is attempt, biological outcome, then housing/saving; computation reverses this order. Ineligible states use the no-attempt branch. **Return to step 2 for the next younger age.** All taste-shock integration is analytic.

4. **Hold policies fixed; construct the stationary cross-section.** In the old stationary economy, start with the entering cohort and propagate its mass forward through all ages using choice probabilities and income, birth, child-departure and survival transitions. Constant entry generates the cross-section. The distant endpoint additionally iterates its household/person demographic fixed point. Aggregate housing policies against household mass to obtain $D(p)$.
5. **Compare demand with supply; change the price and repeat.** Compute the signed relative market error $F(p) = [D(p) - S(p)]/S(p)$. If demand exceeds supply, search toward a higher price; if below, toward a lower one. **At each new price, return to step 1 and re-solve household policies and the distribution.**
The active old initializer brackets the scalar price root and refines it with Brent’s method. The matched endpoint uses a safeguarded bracketed root in log price. A narrow bracket is insufficient: the actual market residual, accounting and feasibility checks must pass; repeat the final solution.

What changes in the simultaneous arm? Replace step 3 by a fertility-nested logsum over complete plans: one housing product for waiting; failure/success products for attempting. Shocks realize jointly before biological success; housing scale cannot exceed fertility scale. Household, distribution and price loops remain.

The transition and calibration algorithms

The stationary calculation is a building block inside two larger loops

A. Solve the perfect-foresight transition at fixed parameters. Households learn the aggregate path in 2007; 2023 is a measurement date on that path. Individual outcomes remain uncertain. Immediate knowledge need not mean an immediate full preference change.

1. **Construct the initial state and solve a stationary endpoint.** Use the old stationary economy for inherited states and pre-2007 cohort history, with observed age/head margins through 2023. Solve the distant stationary economy under terminal preferences, demographics and fiscal rules for V_{T+1} and p_{T+1} . This endpoint is fixed during the price-path iteration.

2. **Guess the entire price vector** $(p_{2007}, p_{2011}, \dots, p_T)$. Here $t + 1$ denotes the next four-year model date. Compute rent from:

$$r_t = (R + \delta + \tau_H)p_t - p_{t+1}.$$

Capital gains affect current rents and choices. Adjust infeasible-rent guesses. The terminal date is numerical closure, not an empirical target year.

3. **Work backward in calendar time.** At T , solve households as on page 1 using next date's values. Move to the preceding model date and repeat back to 2007, storing dated values. All decisions use the same anticipated path.

4. **Start again in 2007 and work forward.** Apply dated policies to inherited mass, measure demand and advance the distribution. Use historical demographic conditioning through 2023 and the person/head law afterward. Never reset each date to a stationary distribution. Recomputing policies from stored values gives two household-solution passes per date.

5. **Update the whole price vector and return to step 2.** Compute $F_t = (D_t - S_t)/S_t$ at every date. A safeguarded Broyden method updates all log prices jointly, initialized from finite-difference price responses. Each trial repeats the full backward/forward evaluation. Require all market errors below 2×10^{-4} (0.02%), other gates and fresh reproduction.

6. **Extend the horizon and check stability.** Compare historical prices/moments and terminal distribution distances. Market clearing can coexist with distortion from an endpoint imposed too early. The stationary continuation value does not reset population to stationarity.

B. Calibrate: place an outer loop around all of A. Let θ contain the 11 free parameters, $\widehat{M}_m$ the empirical moments and M_m the model measurements. The intended joint search minimizes:

$$\widehat{\theta} = \arg \min_{\theta \in \Theta} \sum_{m=1}^{12} w_m \left[M_m(p^*(\theta), \theta) - \widehat{M}_m \right]^2.$$

Here $p^*(\theta)$ must be the equilibrium path for that candidate, not an inherited price guess. The weights include both empirical uncertainty estimates and declared synthetic scales.

1. **Propose θ within its bounds.** The 11 coordinates govern discounting; two fertility-shock scales; the owner-service premium; initial housing-supply scale; two bequest parameters; child housing needs; first-birth cost and housing jump; and the historical preference change. Tenure-shock scale and supply elasticity remain externally fixed.

2. **Renormalize and solve.** Refit the old preference intercept to completed fertility 2.1, rebuild the initial state and endpoint, and run all of algorithm A. Price-root iterations hold θ fixed; calibration iterations change θ .

3. **Measure all 12 targets and score the candidate.** Four fertility moments cover completed fertility, childlessness and two first-birth timing statistics; five housing moments cover the birth response, family-size rooms gap, parent ownership gap, overall ownership and mean rooms; three wealth moments cover wealth/earnings, bequests/wealth and old-age dispersion.

4. **Use parameter sensitivities to propose the next θ ; return to step 2.** Assess rank and weak directions of the 12-by-11 equilibrium-adjusted moment Jacobian. This is different from the date-by-date *price* Jacobian used in A. More targets than parameters does not by itself establish identification.

Dates and current scope. Targets use the 2024 CPS, 1979–84 birth cohorts, 2012–2023 ACS and approved wealth vintages. The birth-housing contrast maps to a 2019–2023 branch. Most model moments refer to 2023. Date/group alignment, horizon certification and new joint calibration remain pending.